

Depth beats breadth: cheap model + adversarial review

0.66× the cost of a premium model — for the same quality

A cheap model (qwen3-235b) run through an adversarial multi-perspective review loop, judged against GPT-5.1's one-shot answer.

WIN

vs. its own one-shot answer

+85%

95% CI +73...+96% · record 42W-5T-1L

WIN

vs. simple self-review

+74%

95% CI +60...+88% · record 37W-10T-1L

TIE

vs. premium model (GPT-5.1)

+12%

95% CI -4...+27% (crosses zero — statistical tie) · record 11W-32T-5L

Net win-rate = (wins – losses) / tasks, 48 hard tasks, blind + order-counterbalanced 3-judge panel.

What is and isn't claimed

Adversarial review on one cheap model reaches premium-quality parity at ~2/3 cost on hard tasks, and massively beats single-shot / simple self-review. It is **not** claimed that review beats premium (the CI crosses zero) — nor that multi-model ensembles help (a separate test found they didn't) — nor that this holds on easy tasks (no headroom to show a difference) or across all future model generations.

Method, every experimental round, and raw numbers: docs/EXPERIMENT_RESULTS.md and docs/PUBLICATION.md — github.com/Kota-Ohno/frugal-fusion